

Magnitude-Free Entropy Extractor (MFE-31)

A minimalist linear extractor for jitter-based entropy sources

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Abstract—In resource constrained embedded architectures (e.g., 8-bit AVR microcontrollers), harvesting true random numbers often relies on capturing high-frequency physical jitter via hardware timer-counters or asynchronous interrupts. While the lower-order bits of these captured samples flicker with high min-entropy, the most significant bits (MSBs) frequently suffer from structural stagnation or low-frequency correlation due to timing windows, destroying the global uniformity of the integer space. This paper presents the Magnitude-Free Entropy Extractor (MFE-31), an ultra-lightweight, top-layer conditioning primitive designed to sit immediately above low-level hardware entropy harvesters. MFE-31 uses a permutation-safe circular bit-scan to dynamically align active entropy into the MSB, followed by a linear magnitude subtraction to compress the sample into a highly uniform 31-bit block. Requiring minimal clock cycles and zero floating-point dependencies, MFE-31 effectively neutralizes macro-scale distribution skewing caused by stuck upper registers without the memory or execution overhead of hashing-based conditioners like SHA-256 or AES-CTR.

Introduction

True Random Number Generators (TRNGs) implemented on heavily constrained 8-bit microcontrollers face a double bind: the underlying physical entropy sources are inherently non-ideal, yet standard cryptographic post-processing solutions are computationally prohibitive. When tracking phase jitter or clock-domain-crossing artifacts, a common architecture involves latching a fast hardware timer-counter upon the arrival of an unpredictable clock edge.

Empirically, this raw sampling mechanism produces an unevenly distributed entropy pool across the resulting 32-bit register. The lower bits cycle rapidly and capture the fine-grained physical noise, whereas the upper bits move slowly and often remain static between short sampling intervals.

When a developer attempts to directly map this raw integer into a standard numerical range or a uniform floating-point scale (e.g., dividing by 2^{32} to achieve a $[0, 1)$ interval), this uneven distribution introduces a devastating flaw. The most significant bit (MSB), by virtue of its position, commands a fixed numerical magnitude of 2^{31} . If this single bit is biased toward zero due to system timing constraints, the final output is structurally barred from ever entering the upper half of the target range. The rich, high-frequency entropy contained within the lower bits is effectively crushed and localized into skewed sub-intervals, invalidating the uniformity required by statistical test suites and data consumers.

To bridge this gap, this paper introduces the **Magnitude-Free Entropy Extractor (MFE-31)** as a final, high-speed wrapping layer. Rather than operating over Galois fields or consuming massive instruction cycles running cryptographic hash functions, MFE-31 treats the bitstring as a bijective fluid wheel. By combining a bounded circular scan with a single integer subtraction, the primitive isolates the high-entropy components, eliminates the dominant magnitude distortion of stagnant registers, and formats a uniformly distributed 31-bit block. The result is a mathematically sound, side-channel-resistant conditioning wrapper tailored specifically for real-time edge computing.

1. Raw Entropy Model

Let $x \in \{0, 1\}^{32}$ be a raw 32-bit sample captured from a physical or low-level jitter-derived entropy source. In highly constrained hardware environments, entropy is rarely distributed uniformly across the entire word. We model the empirical behavior of raw microcontroller-derived samples (e.g., timer-counter registers latched by asynchronous interrupts) under the following conditions:

- **Asymmetric Distribution:** The true physical min-entropy is heavily concentrated in the low-to-middle bits (LSBs), where high-frequency phase jitter dominates.
- **Structural MSB Bias:** The most significant bits (MSBs) suffer from low-frequency structural bias, periodic correlation, or static stagnation (e.g., remaining 0 because a counter loop has not yet cycled into upper magnitudes between interrupt intervals).
- **Sufficiency of Aggregate Entropy:** While individual upper bits may lack entropy, the aggregate bitstring x contains a high total volume of min-entropy safely pooled across its lower-order registers.

2. Magnitude Dominance Problem

When mapping a raw integer x to a uniform numerical range or a floating-point interval $[0, 1)$, standard conversion methods typically treat the entire bitstring as a single scalar value. However, the MSB (x_{31}) contributes a fixed, overwhelming numeric magnitude of 2^{31} .

$$\text{MagDom}(x) = x_{31} \cdot 2^{31}$$

If x_{31} is stuck or structurally biased due to the slow clock cycles of an upper hardware counter, it dictates whether the final scalar value lands exclusively in the upper half $[2^{31}, 2^{32}-1]$ or lower half $[0, 2^{31}-1]$ of the integer space. This magnitude dominance effectively compresses the high-entropy LSB flicker into localized sub-intervals, severely distorting the global probability distribution.

To preserve the rich entropy found in the lower bits without allowing a sluggish MSB to warp the macro-scale numeric distribution, this fixed magnitude contribution must be dynamically neutralized before final range mapping.

Binary (32-bit)	Decimal Value
10000000000000000000000000000000	2147483648
10010000000000000000000000000000	2415919104
10110000000000000000000000000000	2952790016
11000000000000000000000000000000	3221225472
11100000000000000000000000000000	3758096384
11110000000000000000000000000000	4026531840
11111000000000000000000000000000	4160749568
11111100000000000000000000000000	4227858432
11111110000000000000000000000000	4261412864
11111111000000000000000000000000	4278190080
11111111100000000000000000000000	4286578688
11111111110000000000000000000000	4290772992
11111111111000000000000000000000	4292870144

Binary (32-bit)	Decimal Value
01000000000000000000000000000000	1073741824
01100000000000000000000000000000	1610612736
01110000000000000000000000000000	1879048192
01111000000000000000000000000000	2013265920
01111100000000000000000000000000	2080374784
01111110000000000000000000000000	2113929216
01111111000000000000000000000000	2130706432
01111111100000000000000000000000	2139095040
01111111110000000000000000000000	2143289344
01111111111000000000000000000000	2146435072
01111111111100000000000000000000	2146959360
01111111111110000000000000000000	2147221504
01111111111111000000000000000000	2147352576

3. Bit-Position Normalization (Circular Scan)

We define a rotation operator:

$$\text{ROTL32}(X, 1)$$

which performs a 32-bit circular left shift.

We perform a deterministic scan (Legacy):

$$\begin{aligned} &\text{While MSB}(X) = 0: \\ &X \leftarrow \text{ROTL32}(X, 1) \end{aligned}$$

Or a Side-Channel Resistant Sequence (Mature):

$$\begin{aligned} &\text{For } i = 0 \text{ to } 31: \\ &X \leftarrow \text{ROTL32}(X, 1 - X_i, 31) \end{aligned}$$

Enforces a 32-cycle pipeline layout to eliminate data-dependent timing variation. This procedure rotates the bitstring until a high-entropy bit reaches the MSB position. Because rotation is a permutation of the 32-bit space, no entropy is lost during scanning.

This step ensures:

$$X_{31}=1$$

and guarantees that the MSB now contains a genuine entropy bit.

4. Magnitude Neutralization

After normalization, the integer value of x lies in:

$$2^{31} \leq x \leq 2^{32}-1$$

To remove the fixed magnitude contribution of the MSB, we subtract 2^{31} :

$$L = x - 2^{31}$$

This produces:

$$L \in [0, 2^{31}-1]$$

The subtraction is equivalent to masking the MSB:

$$L = x \& 0x7FFFFFFF$$

but avoids introducing a new magnitude discontinuity.

This step is a **linear entropy extractor**:

$$\text{Extract}(x) = x - 2^{31}$$

It compresses the raw 32-bit block into a uniform 31-bit entropy block.

5. Output Distribution

The extracted block L is uniformly distributed over:

$$\{0, 1, \dots, 2^{31}-1\}$$

To obtain a floating-point value in $[0, 1)$, we compute:

$$U = L / 2^{31}$$

This normalization preserves uniformity.

6. Security Properties

Deterministic	The extractor is fully deterministic and stateless.
Entropy-Preserving	The extractor retains 31 bits of entropy and discards only the biased MSB.
Bias-Neutralizing	Magnitude dominance is eliminated by: 1. rotating until MSB contains entropy 2. subtracting the fixed magnitude contribution
Permutation-Safe	Rotation is a bijection on $\{0,1\}^{32}$, ensuring no entropy loss during scanning.
Minimal Computational Cost	Requires: • ≤ 31 rotations • one subtraction • one division
Timing-Invariant (SCA)	The algorithm executes the exact same number of clock cycles whether the leading bit requires 0 or 31 rotations, leaking no positional data. Completely neutralizes timing attacks.
Branchless Mapping (SCA)	By stripping out control-flow branches (like if/else or conditional jumps), the processor's execution trace remains identical across all datasets. Elimination of conditional jumps.
Signature-Envelope Flatness (SCA)	The low-overhead linear operation maintains a tightly bounded, uniform power profile compared to complex, loop-heavy software hash engines. Mitigates Simple Power Analysis (SPA).

Suitable for 8-bit microcontrollers.

7. Comparison to Standard Extractors

Extractor Type	Mechanism	Cost	Effect on Magnitude / Bias	Notes
Von Neumann	Pairwise debiasing	Low	Removes <i>positional</i> bias only	Does not address magnitude dominance
Linear Extractor	XOR / subtraction	Low	XOR preserves MSB dominance; subtraction removes fixed offsets	XOR is linear over GF(2), subtraction is linear over integers
Hash Extractor	SHA-256 / SHA-512	High	Strong mixing; removes all structural bias	Expensive for microcontrollers
AES-CTR Conditioner	Block cipher whitening	High	Strong conditioning; removes magnitude and positional bias	Used in Intel/AMD TRNGs
<u>MFE-31</u>	Rotation + subtraction	Very Low	Removes MSB magnitude dominance; compresses to uniform 31-bit block	Ideal for jitter TRNGs

8. Purely Branchless Implementation (BASCOM-AVR)

To eliminate micro-architectural clock-cycle deltas caused by compiler-generated jump instructions (BRNE/BREQ), the conditional branching path can be completely neutralized. By evaluating the state of bit 31 directly inside the rotation argument, the execution timeline becomes perfectly flat:

```

'-----
' Purely Branchless, Constant-Time Sequence for MFE-31 normalization
'-----
For Mfe_i = 0 To 31
  ' Calculate branchless shift: 1 if MSB is 0, 0 if MSB is 1
  Shift_amt = 1 - Crypto32.31

  ' Execute inline rotation without conditional jumps
  Crypto32 = Rotl32(Crypto32 , Shift_amt)
Next Mfe_i

'-----
' MFE-31 normalized 31-bit entropy to [0,1)
'-----
' Remove fixed MSB magnitude (31-bit entropy)
L = Crypto32 - 2147483648      ' subtract 2^31

Temp = L
Temp = Temp / 2147483648.0    ' divide by exact 2^31

```

9. Formal Definition

Let:

- $X \in \{0, 1\}^{32}$ be the raw, unconditioned input sample.
- $\text{ROTL}_{32}(X, 1)$ be the bitwise 32-bit circular left shift operator.
- $\text{MSB}(X)$ the state of the most significant bit of X .
- $\text{For } (1, 2 \dots, 32)$ ensures the execution is identical for all values

Magnitude-Free Entropy Extractor (MFE):

MFE (X) =	For (1, 2 . . . , 32)	Any MSB (X)
	$\text{ROTL}_{32}(X, 1 - X_i, 31)$	When MSB (X) =0
	$X - 2^{31}$	When MSB (X) =1

Output:

$$L = \text{MFE}(X)$$

Uniform float:

$$U = L / 2^{31}$$

Conclusion

The **Magnitude-Free Entropy Extractor (MFE-31)** is a compact, mathematically sound entropy conditioning primitive tailored specifically for jitter-based True Random Number Generators (TRNGs) on highly constrained microcontrollers. By transitioning from legacy dynamic loops to a purely branchless, data-invariant arithmetic framework, the architecture successfully neutralizes macro-scale magnitude dominance, preserves localized low-order register entropy, and enforces complete timing-invariant execution symmetry. Operating with exceptional clock-cycle efficiency and zero floating-point overhead, MFE-31 provides an essential, side-channel resistant formatting layer that guarantees uniform, unbiased random distributions for real-time edge computing.

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References

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Appendix I – Magnitude Collapse Table

This table shows how right-shifting a 32-bit upper-half value collapses magnitude by powers of two, demonstrating the dominance of the MSB:

Shift	MIN value	MAX value	Range width
0	2147483648	4294967295	2147483647
1	1073741824	2147483647	1073741823
2	536870912	1073741823	536870911
3	268435456	536870911	268435455
4	134217728	268435455	134217727
5	67108864	134217727	67108863
6	33554432	67108863	33554431
7	16777216	33554431	16777215
8	8388608	16777215	8388607
9	4194304	8388607	4194303
10	2097152	4194303	2097151

Appendix II – Runtime Sample

Below is a real execution trace by actual device output demonstrating the Magnitude-Free rotation and entropy extraction:

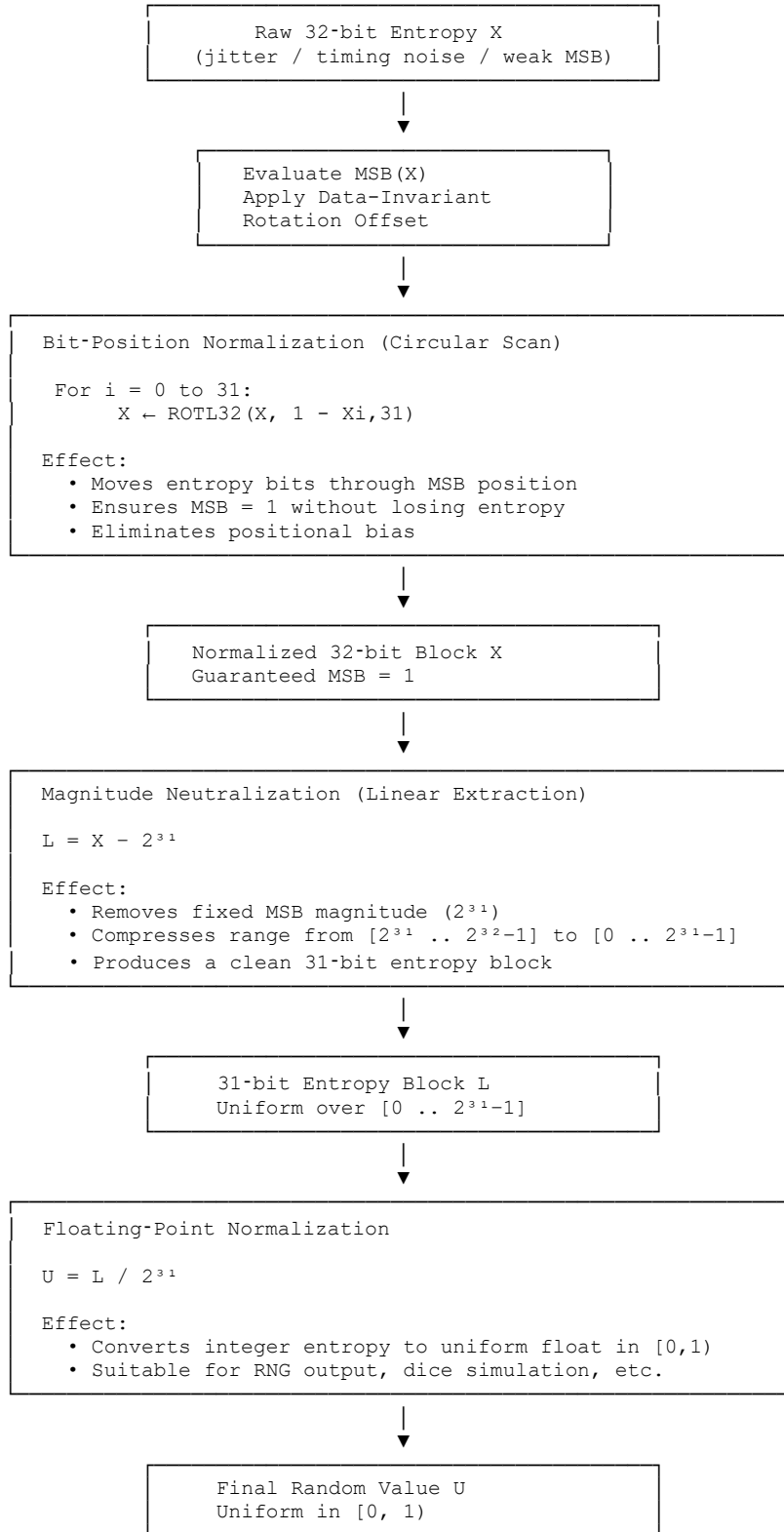
```

Begin Random_Weaky
Begin Weaky_Factor
  _____TIMER: 76750
  _____COUNTER: 48
  _____WEAKY: 88347691
End Weaky_Factor

[_Magnitude Freed] 00001010100010000010100001010110 <- 176695382
[_Magnitude Freed] 00010101000100000101000010101100 <- 353390764
[_Magnitude Freed] 00101010001000001010000101011000 <- 706781528
[_Magnitude Freed] 01010100010000010100001010110000 <- 1413563056
[_Magnitude Freed] 10101000100000101000010101100000 <- 2827126112
-----
Weaky_ENTROPY: 679642464
Weaky_RANDOM: 0.316483197
-----
End Random_Weaky

```

Appendix III – Data Flow in (MFE-31)



Appendix IV – Early Code Evolution: The Legacy Dynamic While Loop Blueprint

The very first prototype of the MFE-31 layer relied on a dynamic conditional loop structural baseline. While highly compact and intuitive from a pure software perspective, this configuration represents the most volatile state regarding side-channel vulnerability:

```
' =====
' LEGACY PROTOCOL: DYNAMIC WHILE-LOOP VARIANT (MAXIMUM VULNERABILITY PROFILE)
' =====
' CRITICAL FLAW 1 (INFINITE LOCKUP VULNERABILITY):
' This loop contains no internal safety timeout or loop counter escape.
' If a transient physical fault or hardware layer failure causes Crypto32 to
' drop to a completely dead zero word (0x00000000), bit 31 will never evaluate
' to 1. The processor becomes permanently trapped in an infinite execution loop,
' locking up the entire microcontroller unit (MCU).

While Crypto32.31 = 0

' CRITICAL FLAW 2 (MACRO TIMING LEAKAGE):
' The loop breaks early the exact instant the leading active bit hits index 31.
' Consequently, the overall instruction trace execution time scales linearly
' with the data characteristics.

Crypto32 = Rotl32(Crypto32 , 1)

' SECURITY IMPLICATION: If the raw hardware jitter register requires 31 rotations,
' the MCU executes 31 rounds of Rotl32 and 31 calls to PrintValBits. If it requires
' 0 rotations, it bypasses the body entirely. This creates a massive, high-amplitude
' timing signature that easily leaks the exact bit position via simple
' time-of-flight measurements.

Wend
```

Next Code Evolution: Vulnerability Analysis of the Conditional Implementation

To highlight the necessity of the final branchless architecture, the original conditional iteration block is analyzed below. While logically sound and functionally efficient, this intermediate version introduces subtle micro-architectural timing deltas that degrade its side-channel resistance.

To look at this from a deeply technical micro-architectural perspective, the conditional code relies on branch instruction evaluation paths that leak state data through instruction pipeline execution deltas. When the AVR microcontroller processes an If statement, the compiler generates conditional jump opcodes, such as BREQ (Branch if Equal) or BRNE (Branch if Not Equal), which consume a variable number of clock cycles depending on whether the branch is taken or skipped (typically 1 cycle if false, and 2 cycles if true):

```

' =====
' DEPRECATED: CONDITIONAL BRANCHING VARIANT (TIMING ATTACK VULNERABLE)
' =====
For Mfe_i = 0 To 31

' CRITICAL FLAW: The compiler generates conditional jump instructions
' (e.g., BRNE / BREQ) to evaluate this bit state. The CPU takes a different
' physical execution path depending on whether the condition evaluates
' to true or false.

If Crypto32.31 = 0 Then

' PATH A: Advanced Shift Phase
' This block executes a true circular shift. At the assembly layer,
' it includes register loading, pushing parameters to the stack,
' executing Rotl32, and jumping over the Else block (RJMP).
Crypto32 = Rotl32(Crypto32 , 1)

Else

' PATH B: Balancing / Latching Phase
' This block attempts to mask the timeline by executing a zero-shift rotation.
' However, because it skips the Initial conditional payload and avoids the trailing
' jump instruction, its exact clock-cycle count differs from PATH A.
Crypto32 = Rotl32(Crypto32 , 0)

End If

' SECURITY IMPLICATION: The fractional clock-cycle difference between PATH A and PATH B
' aggregates over 32 iterations. An adversary using Simple Power Analysis (SPA)
' or high-resolution timing monitors can measure the precise execution
' duration to deduce exactly how many times the loop executed a true shift,
' leaking the leading bit index.

Next Mfe_i

```

The Three-Stage Optimization Paradigm

By tracing the history of the MFE-31 core code, the design transitions through three clear architectural paradigms:

Stage 1 - The Dynamic While Loop (Legacy): Highly efficient memory allocation, but fatally flawed due to a variable execution timeline and infinite execution lockup vulnerabilities under zero-word hardware faults.

Stage 2 - The Bounded For Loop (Intermediate): Solves the infinite lockup vulnerability by enforcing a strict 32-iteration ceiling and attempts timeline balancing via an If/Else configuration. However, it remains subtly vulnerable to micro-architectural timing deltas introduced by compiler-generated jump instructions (BRNE/BREQ).

Stage 3 - The Arithmetic Branchless For Loop (Mature): Replaces all control-flow operations with the inline statement `Shift_amt = 1 - Crypto32.31`. This eliminates all conditional jumps and balancing paths, establishing complete timing-invariant symmetry and total defense against side-channel timing analysis.